Neurath and the Encyclopaedic project of unity of science

by

Olga Pombo

Abstract: Underlying the arguments I will try to put forward, there is a claim and a hypothesis. **My claim:** Neurath's encyclopaedism deserves to be though out as an important and specific program inside neopositivism. We know that neopositivist program for unity of science was not homogeneous, that there are three slightly yet important perspectives needing to be though out in its specific difference: Carnap corresponding to the logical development, Morris to the more pragmatic tendency, and Neurath performing the encyclopaedic trend. **My hypothesis:** Neurath's encyclopaedism is of great significance for the formulation of a contemporary conception of unity of science. A statement which assumes that nowadays encyclopaedia becomes more and more the very model of science and thus that our actuality is giving reason to Neurath.

1. Encyclopaedia as metaphor

Neurath says repeatedly that Encyclopaedia is the model of unity of science, "the model of men's knowledge (...), "the genuine model of science as a whole" (Neurath, 1938: 20), "the soil in which science lives" (Neurath, 1936: 201), "the symbol of the developed scientific cooperation and unity of science" (*ibid*), an image of "our knowledge taken in it totality" (Neurath, 1936: 199).

In a first sense, encyclopaedia is thus for Neurath a **metaphor** of unity of science. A metaphor like many others Neurath likes to use: mosaic, orchestration, sailor, onion. Something which is surprising in an author like Neurath who looks for univocal designation. We know that a metaphor is an expression which, behind its common sense, is able to analogically designate a plurality of other meanings. How to understand its importance for an author like Neurath who is a reformer of language, someone who even proposes an *index verborum* in order to inhibit and even to forbid the use of dangerous words. Of course we know that by dangerous words Neurath meant just those words which avoid communication. And that is not the case of metaphor which, on the contrary, is a strong communicative linguistic procedure able to make us *see* what the concept means, in this case, the concept of unity of science. Almost an *Isotype*, a pictorial sign, a hieroglyphic which condenses dense information and exhibits it through its spatial form. We must remember that Neurath is also a constructor of a new language endowed with common intelligibility and great communicative capacity.

In this respect, it is interesting to make two observations:

1. Unity of science always gave rise to strong metaphors. That is the case of the **circle**. From Antiquity (from I a/c. Cicero' s circle of liberal arts to V d.C. Martianus Capella's *disciplinae Ciclicae*, in <u>Nuptiis Philosophiae et Mercurio</u>) until Hegel, Adler or Piaget, the circle has been the metaphor of eternity, divine perfection, stability, systematicity, immobility, closeness, no beginning, no end, no hierarchy, a metaphor by which unity is theologically thought out. That is

also the case of the **tree** which, from Rámon Lull, to Bacon and Descartes represented a dynamic form, a living being, an organic development, historicity, generation, multiplicity, vertical subdivision, mostly dichotomic al. Unity is there thought out as hierarchical. That is also the case of Diderot and D'Alembert's metaphor of the **Mapamundis** by which unity of science is thought as a topographical, topological, cartographic, territorial, juxtaposition. Unity is the horizontal result of the complementary work of different sciences, of its ordered evolution and cumulativity, or yet Leibniz's metaphor **Oceanus** as permanent connection, fluidity. That is also the case of Couvier, Comte, Oppenheimer's metaphor of the **House** by which unity of science is thought out as architectonic, previously planned assemblage of elements, growing by savage proliferation[1].

2. Neurath never uses any of the metaphors proposed by his antecessors. He, always so devoted to praise the achievements of the past, he proposes a new metaphor – encyclopaedia - a word which, in itself, is already a metaphor coming from the Greek *eu kuklios paideia*, the perfect circle or complete course of knowledge and education.

What does it mean that metaphor? That is what we will try to understand. However, before that, let us just remember that encyclopaedia is not a metaphor as many others Neurath uses to mean unity of science. Encyclopaedia is a Neurath's **idea** in origin[2], a **project** first conceived by Neurath and in which he works since 1920 up until the end of his life, a project afterwards proposed, discussed and approved in the First International Congress for the Unity of Science in 1935 but of which Neurath seems to have first talked with Einstein and Hans Hann and only afterwards with Carnap and Philipe Frank; it is an **attitude**, an open, co-operative and anti-dogmatic "attitude" (cf. Neurath, 1937: 141); it is "**plataform which makes it possible** to find out how much cooperation is in fact possible" (Neurath, 1937:137); it is the "**simbol** of a developed scientific co-operation, of the unity of sciences and of the fraternity among the new encyclopaedits" (Neurath, 1936: 201); it is a **program's life** for "men of good will" (Neurath, 1936: 200). As Neurath says, "it is the very practice of life which imposes encyclopaedic task" (Neurath, 1936: 199). Encyclopaedia is also a **movement**, a cooperative endeavor by which Neurath tried to bring "together scientists in different fields and in different countries, as well as persons who have some interest in science or hope that science will help to ameliorate personal and social life" (Neurath, 1938: 1). Practical thinker par excellence, Neurath was aware of the need of overcoming "dreams by acts" (Neurath, 1936: 200), of giving institutional form to theoretical ideas. With all his energy and perseverance, he dedicated himself to the social engineering of science organizer, building the necessary instruments – congresses, institutes, museums, series of books, associations, education, journals and – of course – that powerful mechanism of unification of knowledge which is encyclopaedia.

So, encyclopaedia is in Neurath not only a **metaphor** but also a **realization**, a concrete, **material work** which, even if far from the ambitious plan, he succeeded to put forward, overcoming a set of big difficulties.

A work which Neurath puts in the line of a long history of similar endeavors: the cosmic poems of antiquity, the *Sumae* of Thomas Aquinas, the *Ars Magna* and *Ars Generali* (1308) of Ramon Lull, the *Instauratio Magna* (1620) of Bacon, Comenius's *Pansophia*, *De Rerum Humanorum Emendatione Consultatio Catholica* (1642-1670), Leibniz's *Encyclopaedia sive Sciencia Universalis and Atlas Universalis*, Hegel's *Encyclopädie der Philosophischen*

Wissenschaften (1817), Schelling's *Naturphilosophie*, A. Comte's *Cours de Philosophie Positive* (1830-1842), Spencer's *Synthetic Philosophy* and, of course, the French *Encyclopédie* of Diderot and d'Alembert (1751-1765).

Three remarks must be made.

First, Neurath only refers philosophical encyclopaedias. He could have mentioned the *Encyclopaedia Britannica* (1st edition 1768-1771), the *Encyclopaedia Metropolitane* by Coleridge (1817-1845), *Le Grand Dictionnaire Universel de Larousse* (1866-1890) or the monumental *Allgemeine Encyclopädie der Wissenschaft* (1818-1889) of Ersch and Grüber (167 volumes). But he did not. And he did not because he knew that the aim of a philosophical encyclopaedia is less to give an exhaustive exposition of the totality of knowledge (empirical, scientific and technical) than to stress the articulation, the integration, the connections, the universal relations of the various kinds of knowledge. In a word, to make us remember that knowledge has a unity.

Secondly, the list of philosophic encyclopedias Neurath gives is quite exhaustive. It includes inclusively some attempts previous to the XVII century, the time in which encyclopaedia reaches its very identity. Of course he forgets some: he forgets Alsted's *Encyclopaedia Omnium Scientiarum* of 1630; he forgets Novalis's *Das Allgemeine Brouillon*, a posthumous publication of 1802; he forgets St. Simon, whose *Prospectus à une Nouvelle Encyclopédie* (1810) constitutes an important formulation of a theoretical program stressing the need to articulate all the domains (science, politics, social life) according to the scientific attitude and methodology proposed by Bacon and realized by Newton in natural sciences. Something which Neurath would appreciate.

In third place, of those philosophic encyclopedias he quotes, some are just labels, references, celebrated ancestors (like Lull, Comenius, Spencer), others are enemies, adversaries, projects with which Neurath wants to establish strong opposition (Thomas Aquinas, Hegel)[3]. Others are recognized like major predecessors (Bacon, Leibniz). Finally, it is the Encyclopédie of Diderot and D'Alembert, which is claimed to be the great inspiration. The very introduction Neurath writes for the new Encyclopaedia of Unified Science – the text untitled "*Unified science as Encyclopaedic Integration*" (1938) – is clearly marked by such illustrious ancestor which is the Discours Préliminaire (1751) of D'Alembert. As Neurath writes: "Our encyclopaedia continues the famous French *Encyclopédie*" (Neurath, 1938: 2). And further, "About one hundred and ninety years ago, D'Alembert wrote a Discours Préliminaire for the French Encyclopédie, a gigantic work achieved by the co-operation of a great many specialists (…). One must carefully look at their work as an important example of organized co-operation" (*ibid*)

2. Encyclopaedia main determinations

Let us now see what is common to all these projects and how is Neurath positioned face to the main determinations of the encyclopaedia. We will try to briefly characterize encyclopaedic project in eight points.

1. Encyclopaedia aims to become a complete, impartial and objective reflection of all knowledge conquered by mankind and available at a certain historical moment. This vertigo towards exhaustivity can lead encyclopaedia to a teratological dimension – the case of the immense Chinese encyclopaedias which Neurath refers is eloquent – the Yung-loh, XV century, 11.995 volumes (never ended), the Tu shu chi ch'eng published

in Shanghai, 1726, with its 5.020 volumes and of which there is a complete exemplar at the British Museum.

However, in the Western world, encyclopaedia is touched by the law of constant innovation that characterizes our civilization. It is thus always designed, not as a *complete* but as a *compact* library (the aim of encyclopaedia is to put library inside the book), an economic work forced to combine exhaustivity with selectivity. In the line of Bacon's indications, encyclopaedia is assumed as an historical production, always unfinished, incomplete, precarious, and condemned to the voracity of knowledge progress[4].

This is what Neurath recognizes when he stresses that "encyclopaedia is a provisional assemblage of knowledge, not something incomplete but the reunion of scientific knowledge which we possess at present" (Neurath, 1936: 188). I quote again: "The future will produce new encyclopaedias (…) and it is senseless to speak about a complete encyclopaedia" (*ibid*). And further, "the progress of science goes from one encyclopaedia to another one" (*ibid*). Like science, encyclopaedia is something to go on doing "little by little" (Neurath, 1938: 3). Sciences are living realities. Reductionism is an infinite task.

2. Encyclopaedia is not a dictionary. Dictionaries aim to be a complete codification of language, even if they can never realize such a project and thus they all suppose some encyclopaedic openness to the world. On the contrary, encyclopaedia is a semantically opened structure, a representation referring the world of things and events which are to be spoken, that is, to be known. Even if some encyclopaedias may have been designated as dictionaries (the most celebrated example if the Encyclopédie ou Dictionnaire Raisonné des Sciences, des Arts et de Métiers, by Diderot and d'Alembert or Larrousse's Grand Dictionnaire); even if some

have in common with the dictionaries the alphabetic presentation of its elements (the case again of Diderot and d'Alembert's Encyclopédie or of Coleridge's Encyclopaedia Metropolitana) – encyclopaedia is never a dictionary. Encyclopaedia is not interested in words but in what words mean and refer – the world behind the words.

That is why encyclopaedia always reflects the cultural and scientific situation in which it is created. That is why encyclopaedia needs constant actualizations. As it is said in the Preface of the 1974 edition of the Britannica, "if we want to seriously reflect the knowledge situation of nowadays, we cannot dedicate 30 pages to chivalry and 31 to legal status concerning pornography" (Adler, 1974, vol. I: XIII). And to actualize it is not just to add new entries (for instance, bioethics) but to diminish the importance of some (for instance, flogistus) and to grow up others (for instance, atom). As Neurath says: "the encyclopaedia which we aim is a given historical formation to which any extra-historical ideal can be opposed" (Neurath, 1936: 200). And he adds: "Our attempt can obviously be rejected later. That is something we must be aware. Nevertheless, the new generation will probably improve the work with enthusiasm and success towards unitary science." (Neurath, 1936: 200-201)

3. If encyclopaedia is never a dictionary, yet they have one point in common. Like dictionary, encyclopaedia is a discontinuous text made of independent segments or *entries*, either alphabetically organized or structured in larger conceptual, thematic or disciplinary frameworks.

Those semantic fields never present well-defined borders. Each entry opens (explicitly or implicitly) to other entries which, in turn open to others, in such a way that each entry is virtually connected with all others. In other words, encyclopaedia is not so much a monumental reunion of all knowledge in one closed place, but the free circulation of unity throughout the dense and sensual effectivity of its volumes and pages. It is not a static totality but a dynamic entity, "a living being and not a phantom (…), not a mausoleum or an herbarium, but a living intellectual force" as Otto Neurath said in his famous *Unified Science and Encyclopaedic Integration* (1938: 25-26).

4. The material objectivity of encyclopaedia has thus an unlimited condition. The finite member of its pages contains a net of discrete elements which can be articulated according to multiple relations in an undetermined number of combinations, a kind of combinatory without rule[5].

That is to say, behind the additive synthesis of all its entries, encyclopaedia does point to the exhaustion of all the possible combinations of its entries. That is the point in which encyclopaedia makes unity of science appear more like an infinite task. As Neurath recognizes, the experience of infinite gives great pleasure: "many young people, to whom sciences appear cold and distant in their isolation, will surely be attracted to unified science because of the possibility of connecting everything with everything" (Neurath, 1937:140).

That is why encyclopaedia offers its readers the possibility of making their own journey of reading according to their interests and preferences. In fact, encyclopaedia not only offers that possibility but also suggests it, promotes it, invites the reader to take his own course, proposing a set of resources (for instance, indexes, thesaurus)by which the chooses, by successive extension, which semantic fields he should read after another. That is why the International Encyclopaedia of Unified Science "will not be", as Neurath says, "a series of alphabetically arranged articles but rather a series of monographs with a highly analytical index"

(Neurath, 1938: 24). And, in the very heart of the onion, there will be "two volumes which will deal with the problems of systematization in special sciences and in unified science" (*ibid*).

5. That is why encyclopaedia always has a strong hope in its cultural, educative role.

It is true that, inviting the reader to follow his own *cursus*, the encyclopaedia is not didactical, not a student's manual. The reader is never a student, never a pupil, someone who intends to follow a pre-determined *curriculum* in order to obtain a systematized knowledge. Neither is he an autodidactic – caricature and victim who tries to substitute school by encyclopaedia[6]. The reader of an encyclopaedia is always an already lettered public – "un publique éclairé" as D'Alembert says[7], a "curious and intelligent reader", as stated in the Preface of the Britannica (Adler, 1973-74, vol. I: XV).

However, encyclopaedia always supposes the constitution of a new knowledge community whose sociological limits ideally coincide with the entire humanity. So, in order to reinforce its cultural, educative and even ideological role, encyclopaedia points to the semantic exploration of the diagrammatic resources of language putting them at the service of the iconic and imagetic description of the world. That is why encyclopaedia frequently includes non-linear materials such as pictures, drawings, diagrams, illustrations, maps, statistic lists, plans, and tables of all types (see the case of the more than 600 pictures of the Encyclopédie and of the 11 complementary volumes of pictures Diderot published in 1762-1772).

We know that Neurath stressed, in theory and in practice, the need of democratization and popularization of knowledge. See his many activities in terms of social and political education, as Museum director in Vienna (1925-34), school reformer (in the line of Otto Glöckel social democratic school reformer movement[8]), militantly working in adult education at the *Mundanaum Institute*, and, above all, inventor of the Vienna Method of Picture Statistics and International System of Typographic Picture Education or Isotype. By creating Isotype, Neurath said, "I was thinking mainly of the masses who could now grasp something more than before of the present knowledge of mankind" (Neurath, 1946: 502). It is thus quite understandable that, in what concerns encyclopaedia, Neurath pointed to the construction of a 10 volumes *Isotype Thesaurus* including all kind of pictorial representations able of "showing important facts by means of unified visual aids" (Neurath, 1938: 25) and that he places such a project in the line of Leibniz *Atlas Universalis* and Comenius *Orbis Pictus* (cf. Neurath, 1938: 16). That is to say, Neurath understood well the close connection between Museum and Encyclopaedia – Museum is a material encyclopaedia. Encyclopaedia tends to recover the idea of museum. The both are seeing machines.

6. Further, encyclopaedia is a collective work. It is true some works today can be included in the gender of encyclopaedia were made by one author only. That is the case of Varro (116 b.C. – 27 b.C.) Rerum Divinorum et Humanorum, of Plinius (23-79) Historia Naturalis, of medieval works by Isidorus of Sevilla(c. 560 – 636) Etimologies, Vincent de Beauvais (c. 1190-1264)*Speculum Majus* and those many Renaissance encyclopaedia like Giorgio Valla (De expetendis et fugiendis rebus, 1501)), Rafaele Maffei (Commentarium, 1506), Domenico Delfini (Summario di Tutta Scienza, 1556), Luis Vives (Tradentis Disciplinis, 1531), Comenius (De rerum humanorum emendatione consultatio catholica, 1662-1664), Alsted Encyclopaedia Omnium Scientiarum, 1630) or Pierre Bayle (Dictionaire Historique et Critique, 1697).

But, from XVIII century on, encyclopaedia supposes the collaboration of different competencies: half a dozen of celebrated science men like John Ray and Newton as in the case of John Harris Lexicon Thecnicon (1704); many unknown, unidentified, even anonymous collaborators (like in the case of Diderot's Encyclopédie[9]); various identified authors presenting their controversial perspectives as put in practice in the XX century[10]. As Neurath says, "In these volumes, scientists with different opinions will be given an opportunity to explain their individual ideals in their own formulation" (Neurath 1938: 25); "The collaborators will certainly learn from their encyclopaedical work. Suggestions from different sources will stimulate this activity so that this Encyclopaedia will become a platform for the discussion of all aspects of scientific enterprise" (Neurath 1938:26). As he writes: encyclopaedia "would always be open to questions and give rise to innumerable controversies" (Neurath, 1937: 140). That is, from one's voice discourse, encyclopaedia becomes a plural, pluralistic, polymorphic, democratic, international "orchestra" (cf. Neurath, 1946).

Like all orchestras, it needs a maestro – to do what? To do bring together the various and diverse instruments, to coordinate the differences, to support a "working community" (Neurath, 1937: 137), in a word, to "harmonize" the multiplicity (cf. Neurath, 1946: 498). The fundamental aim is: "The maximum of co-operation. That is the program!" (Neurath, 1938: 24).

7. Collective work, encyclopaedia is never an amount of discontinuous elements coming from different sources. It is never a miscellany, never an inventory, but an ordered presentation. As Leibniz said, "l' encyclopédie est un corps oú les connaissances humanise les plus importants sont rangées par ordre" (Leibniz, Gerhard (ed.), 1960, vol.7: 40). It always

supposes a "systhème figuré des connaissances humaines", a *mapamundus* where the *order* and *connection* of human knowledge can be discovered, as stressed by Diderot and D'Alembert.

Let us say it clearly: encyclopaedia always supposes, implicitly or explicitly, a system of organization of knowledge. This systematization can be disturbed by the thematic or disciplinary order or even concealed (hidden) by the alphabetic presentation of entries. But the systematic structure is there and it is that systematic structure which determines both the quantity and quality of the entries, the inclusion or exclusion of certain topics, the settling, the articulation, the ordering, the relative status and importance of some entries towards other entries.

However, that does not mean that encyclopaedia should be endowed with a systematic perspective, a constraining point of view. Neurath is very strict concerning this point: "any kind of intellectual absolutism should be avoided as not being in harmony with our scientific practice which is of the same type as our everyday life" (Neurath, 1947: 80). "We must, of course, avoid the error of trying to anticipate *the* system as our model of science. Our model is encyclopaedia itself" (Neurath, 1937: 136). "The task of encyclopaedia is to represent the present state of science and not to *anticipate* an unanimity which does not yet exist" (Neurath, 1937: 139. In a much clear formulation, Neurath asserts: "The anticipated completeness of *the* system is opposed to the stressed incompleteness of encyclopaedia" (Neurath, 1938: 21).

Neurath's so claimed heritage face to the French Encyclopédie comes precisely from the fact that Encyclopaedia should constitute a true "alternative to systems" (Neurath, 1938: 7). French Encyclopédie had not the aim, typical of metaphysics, of gaining an absolute point of view, of starting on the most general propositions in order to deduce the particular sciences. Similarly, the Encyclopaedia of logical empiricism should refuse any systematic totalization. It cannot be a unique, definitive whole. It must renounce foundationalism. It must accept the historical, provisory character of all synthesis. That is why it "should continue in some way the work which d'Alembert put forward with his extreme dislike by systems (Neurath, 1936: 201).

That is to say, the synthesis Neurath aims is of Baconian nature, a reunion always provisory and opened of empirically grounded knowledge.

8. Last point. If it is true that encyclopaedia reflects the knowledge situation of its time, in what concerns the organization of knowledge, encyclopaedia has also a prospective role, both in its practical, ideological, political, educative aims and in its high heuristic value. This is the point in which Neurath comes close to of Leibniz encyclopaedism, namely to its most meaningful feature: the heuristic value of encyclopaedia.[11] For Neurath as for Leibniz, encyclopaedia is a kind of an *organon* at the service of science progress and search for the truth.

By establishing cross-connections, by doing "local systematizations" (Neurath, 1946: 498), by promoting "terminological unifications" (Neurath, 1936: 196), by advancing "aggregations" (Neurath, 1936: 188), by developing "transversal connexions" (Neurath, 1936: 197, 198), by showing "the gaps in our present knowledge and the difficulties and discrepancies which are found at present in the various fields of science" (Neurath, 1938: 25), encyclopaedia reveals itself clearly as an *organon* at the service of science progress and search for truth. By overcoming the "speculative juxtaposition" (Neurath, 1938: 20), by the putting together of several disciplines, by taking into practice a co-operative articulation, encyclopaedia, as Neurath says, allows scientists to build up “systematic bridges from science to science, analyzing concepts which are used in different sciences, considering all questions dealing with classification, order, etc” (Neurath, 1938: 18), to establish a “comparison of the argumentation in cosmology, geology, physics, biology, behavioristics (‘psychology’), history and social sciences” (Neurath, 1938: 14), in such a way that "advances in one will bring about advances in the others” (Neurath, 1938: 24).

In a word, by synthesizing the already known, by giving to know what is known, encyclopaedia constitutes a kind of artificial protesis which liberates natural memory for what really matters – the unknown. Encyclopaedia – we could say – empties the opposition between memory and invention. An opposition which can only be thought out upon the disregard of Leibniz intimate connection between the *ars judicandi* and the *ars inveniendi*.

3. Conclusive remarks

We can now approach a modest conclusion. We did recognize Neurath’s project of encyclopaedia in all those 8 issues. And we have to give him reason. We have to recognize that Neurath’s encyclopaedism is of great significance for the formulation of a contemporary conception of unity of science. Maybe we are now in a better position to understand why encyclopaedia is, in fact, the metaphor of unity of science.

Unlike circle, encyclopaedia does not need divine perfection. “It would of course be nice to harmonize the demonstrations in all areas, but in the meantime, scientific research must proceed” (Neurath, 1946: 498). Encyclopaedia does not need eternity, stability, and complete systematicity. We can start with what we have. “If we don’t have a system by the top we can build a system by down” (Neurath, 1936: 196). Utopia in Neurath is not an ideal located far in an impossible future but an active attitude.

Unlike tree, encyclopaedia does not need hierarchy. Encyclopaedia does not believe in “leader’s intuition” (Neurath, 1946: 504), it “challenges any intellectual authority which pretends to preach the truth” (*ibid*), it does not need “transcendental credos” (Neurath, 1946: 505), nor “centralized and dominating zeal which always lead to self-sacrifice and sacrifice of others” (*ibid*). Encyclopaedia demands pluralism, tolerance, perspectivism. It accepts arguing, it works with a great many scientific units of varying magnitude and differing provenance. It starts from everyday language; it rejects absolutism, pyramidism, foundationalism, and substitutes that by fraternity. Unity does not imply the exclusion of variety. On the contrary, it demands it.

Unlike house, encyclopaedia does not need previous planification, no super-science or pseudo-nationalistic anticipation of the system of science. But it must fight against savage proliferation, disciplinary terrorism, closeness of specialties. Encyclopaedia needs organized cooperation, socialist planification. Encyclopaedia is an orchestra (also a good metaphor which Neurath uses for encyclopaedia), plural, controversial, cross connected, a potential multiplicity able to overcome its own limits and capacities. The maestro must be democratic (giving voice to all instruments). That is why encyclopaedia is a so deeply anti-Cartesian endeavor. It is not the work of a meditative singularity. Nor is it grounded in any indisputable truth. It does not make *tabua rasa* of the competencies and virtuosim of the members of orchestra. The task of the maestro is mostly to harmonize.

Unlike mapa mundus (Diderot and d’Alembert preferred metaphor), encyclopaedia does not need previous cartography, no previous classification of sciences. A mere librarian classification is enough[12]. Above all, encyclopaedia does not have any territorial, colonialist, imperialist conception of knowledge. To progress in knowledge, to know more, is not to conquer another foreign country. To know more is to establish new fraternities, new interdisciplinary forum.

That is to say, Neurath's encyclopaedia is close to the celebrated Lebnizian oceanic metaphor[13] for the unity of science: "The entire body of science can be considered as the ocean since it is continuous and without interruption or separation even if men consider in it different parts giving them names according to their commodity" (Leibniz, 1903: 530-531).

Apart from arbitrary, institutional borders, encyclopaedia points to a fluid, infinite, combinatory regime, aiming to promote the free circulation – something which clearly announces the curiosity of navigation[14] in the electronic encyclopaedia and Internet - in the interior body of the encyclopaedia.

That is why "we are like sailors who have to rebuild their boat at open sea, without ever coming to a safe and dry coast and rebuild it on the basis of the best materials"

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[1] We have studied elsewhere the main metaphors of Unity of Science which have been proposed since the Hellenistic circle up until the electronic net (cf. Pombo, 2006)

[2] As Charles Morris says(1969: IX), "Encyclopedia is, in its origin, a Neurath’s idea". See also the testimony Carnap gives in his Intellectual Autobiography (Carnap, 1963: 23)

[3] In this respect, it is interesting to note that Neurath praises A. Comte Philosophie Positive (cf. Neurath, 1938: 8), even if he should blame him almost for the same anti-fundamentalist reasons he rejects Hegel’s piramidism.

[4] As Bacon writes in the Preface to the Instauratio Magna: “it does not suppose that the work can be altogether completed within one generation, but provides for its being taken up by another" (Bacon (1620), vol. IV: 21).

[5] This is one of the major points in which encyclopaedia shows its close relationship with Internet. Cf. Pombo (2006a)

[6] That is the tragedy of the extraordinary work of Flaubert Bouvard et Pécouchet (1880), the unwise adventures of two heroes taken by encyclopaedist passion who succumb the labyrinth of knowledge mostly, we would say, by absence of an ordered plan of studies.

[7] “It can work as a library in all subjects for a man of the world and in all subjects except his own, for a science professional” (D'Alembert, 1965: 143).

[8] On the practical, militant and educational philosophical and encyclopaedical activity of Neurath, cf. Haller (1991) and Stadler (1991).

[9] The Encyclopédie had in fact the colaboration of first level science men, artists, musicians, writers like Quesnay, Rousseau, Voltaire, Du Marsais, Turgot, Montesquieu, Grimm or Duclos, side by side with craftsman, agricultures, gardners, weavers, etc. and even many spontaneous and sometimes anonimous "coleagues", all united by a militant "intéret général du genre humain et par un sentiment de solidarité reciproque" as Diderot says (1994: 368).

[10] Around 4000 in the case of the 15th edition of the Britannica (cf. Adler, (1974), Preface, vol. I: XVIII)

[11] As Leibniz states, "le principal est que la reveue exacte de ce que nous avons acquis faciliteroit merveilleusement des nouveaux acquest" (GP 7: 159). For further developments on Leibniz encyclopeadism, cf. Pombo (2002)

[12] In this respect, see a curious passage by Neurath in his article *The Departmentalization of Unified Science* (1937a) in which the opposition to what he calls there as "pyramidism" is enlarged to the anticipative models of science classification. As Neurath writes: "Pyramidism (...) intends to built a symmetrical and complete edifice of the sciences by means of main divisions, subdivisions, subsubdivisions, etc" (Neurath, 1937a: 245). And ads: "Encyclopaedism isd satisfied with a rough bibliographic order for an initial orientation, made by librarians" (*ibid*).

[13] Neurath himself recognizes the oceanic character of encyclopaedism even if, quite surprisingly, he relates it not with Leibniz but with Freud. As he asks: "Is such a pure scientific Encyclopaedism in a position to satisfy human yearnings and to create 'oceanic feeling' – if we may use Freud's term in this case? This question will be answered by the Man of the Future" (Neurath, 1938b: 484)

[14] The concept of "navigation", appears explictly at the Organon of the Enciclopaedia Universalis , vol. XVII: 595. Encyclopaedia Universalis, Symposium, Paris: Encyclopaedia Universalis France S.A.